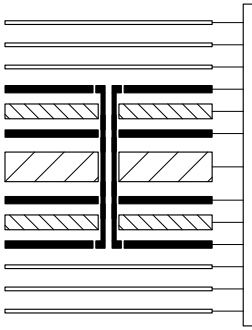
# CanSat Devboard Fabrication Document

## Layer Stack Legend



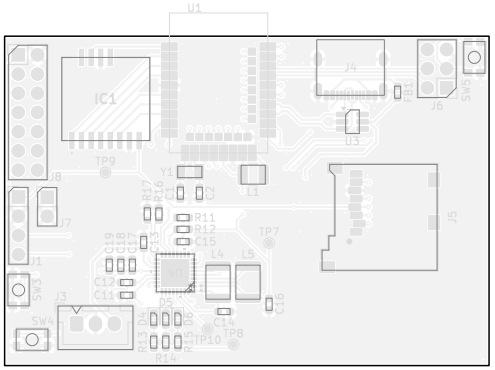
| Material | Layer | Thickness | Dielectric | Type | Gerber |
|---|---|---|---|---|---|
| | F.Paste | | | Paste Mask | |
| | F.Silkscreen | | Direct Printing | Legend | GBR |
| | F.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| **Copper** | **L1 (Sig, PWR)** | **0.07mm (2.00oz)** | | **Signal** | **GBR** |
| Prepreg | | 0.1mm | FR4_7628 | Dielectric | |
| **Copper** | **L2 (GND)** | **0.035mm (1.00oz)** | | **Plane** | **GBR** |
| Core | | 1.21mm | FR4 | Dielectric | |
| **Copper** | **L3 (GND)** | **0.035mm (1.00oz)** | | **Plane** | **GBR** |
| Prepreg | | 0.1mm | FR4_7628 | Dielectric | |
| **Copper** | **L4 (Sig, PWR)** | **0.07mm (2.00oz)** | | **Signal** | **GBR** |
| | B.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| | B.Silkscreen | | Direct Printing | Legend | GBR |
| | B.Paste | | | Paste Mask | |

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating

## Impedance Table

| Transmission Line | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|---|---|---|---|---|---|---|
| Edge-Coupled Coated Microstrip | 100 | ±10 % | L1 | 0.2032 | 0.28 | L2 |

## Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY KARPAT SPACE PROGRAM.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

| | |
|---|---|
| BOARD SIZE | 64.059 × 43.561 mm |
| BOARD THICKNESS | 1.660 mm |
| TRACE WIDTH | 0.200 mm |
| TRACE TO TRACE | 0.200 mm |
| MIN. HOLE (PTH) | 0.250 mm |
| MIN. HOLE (NPTH) | 0.650 mm |
| ANNULAR RING | 0.150 mm |
| COPPER TO HOLE | 0.250 mm |
| COPPER TO EDGE | 0.250 mm |
| HOLE TO HOLE | 0.100 mm |

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

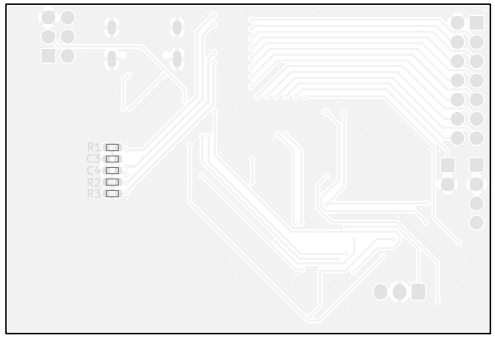
12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: Top Fabrication (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 1 of 10 |

# CanSat Devboard Fabrication Document

## Bottom Fabrication (Scale 1:1)




All dimensions are in millimeters unless otherwise specified.

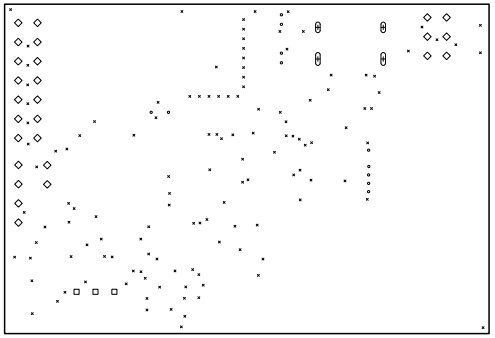
| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: Bottom Fabrication (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 2 of 10 |

# CanSat Devboard Fabrication Document

## Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 130 | 0.25mm (9.84mils) | PTH | Round | L1 (Sig, PWR) - L4 (Sig, PWR) | Via |
| ○ | 11 | 0.30mm (11.81mils) | PTH | Round | L1 (Sig, PWR) - L4 (Sig, PWR) | Via |
| + | 4 | 0.60mm (23.62mils) | PTH | Slot | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad |
| □ | 3 | 0.95mm (37.40mils) | PTH | Round | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad |
| ◇ | 26 | 1.00mm (39.37mils) | PTH | Round | L1 (Sig, PWR) - L4 (Sig, PWR) | Pad |
| | Total 174 | | | | | |

## Drill Drawing L1 - L4 (Scale 1:1)



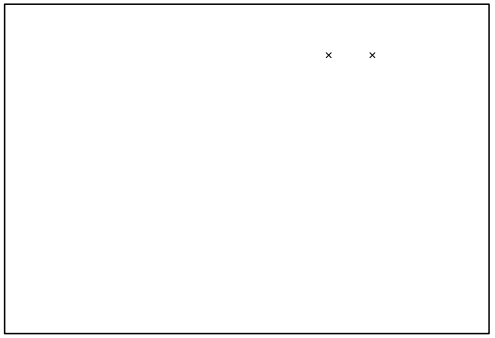
| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: Drill Drawing (L1 - L4) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 3 of 10 |

# CanSat Devboard Fabrication Document

## Drill Table

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 2 | 0.65mm (25.59mils) | NPTH | Round | L1 (Sig, PWR) - L4 (Sig, PWR) | Mechanical |
| | Total 2 | | | | | |

## Drill Drawing L1 - L4 (Scale 1:1)



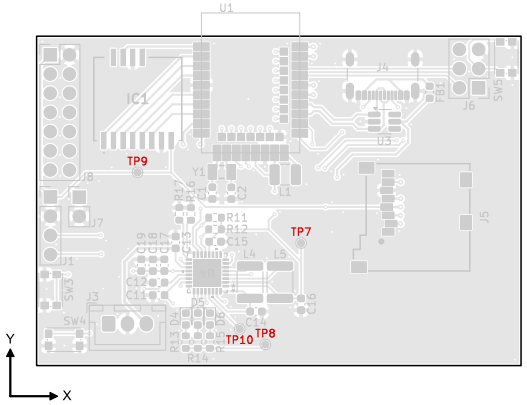
| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: Drill Drawing (L1 - L4) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 4 of 10 |

# CanSat Devboard Fabrication Document

## Top Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|
| TP7 | VMCU | 87.96 | 1.25 |
| TP8 | VSYS | 83.23 | -12.16 |
| TP9 | +3.3V | 66.31 | 10.60 |
| TP10 | VLDO | 79.83 | -10.12 |

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|



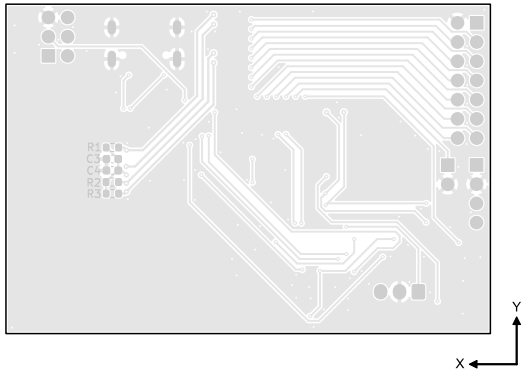

All dimensions are in millimeters unless otherwise specified.

| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: Top Test Points (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 5 of 10 |

# CanSat Devboard Fabrication Document

## Bottom Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|

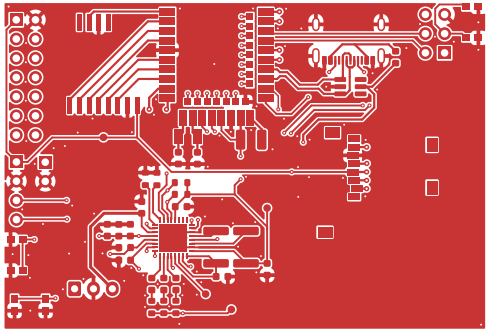



All dimensions are in millimeters unless otherwise specified.

| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: Bottom Test Points (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 6 of 10 |

# CanSat Devboard Fabrication Document

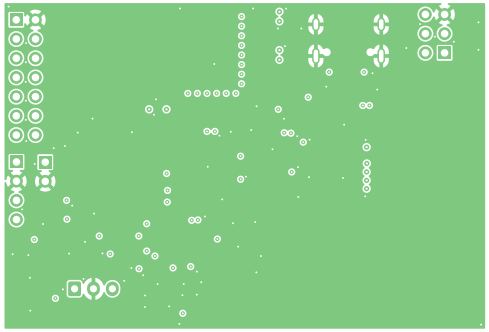
## L1 (Sig, PWR) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| Comments: | Company: **Karpat Space Program** | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: **CanSat Devboard** | | Project Name: **Cansat Devboard** | |
| Sheet Title: L1 (Sig, PWR) (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: **A4** | Sheet: **7** of **10** |

# CanSat Devboard Fabrication Document

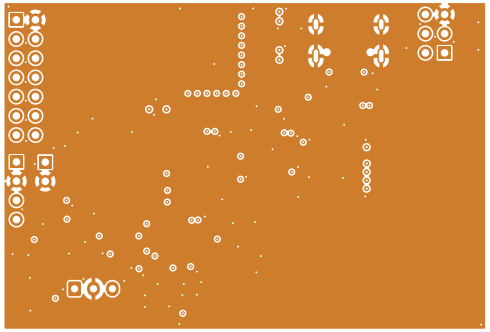
## L2 (GND) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: L2 (GND) (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 8 of 10 |

# CanSat Devboard Fabrication Document

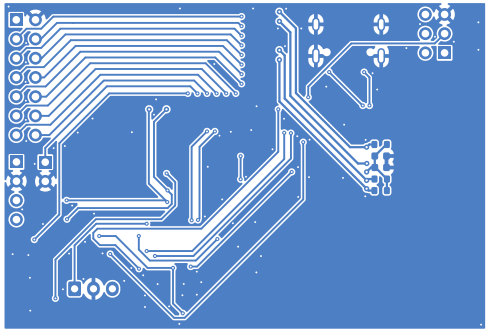
## L3 (GND) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| Comments: | Company: Karpat Space Program | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: CanSat Devboard | | Project Name: Cansat Devboard | |
| Sheet Title: L3 (GND) (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: A4 | Sheet: 9 of 10 |

# CanSat Devboard Fabrication Document

## L4 (Sig, PWR) (Scale 1:1)



| | | | | |
|---|---|---|---|---|
| Comments: | Company: **Karpat Space Program** | | Variant: PRELIMINARY | Git Hash: f0175fe |
| | Board Name: **CanSat Devboard** | | Project Name: **Cansat Devboard** | |
| Sheet Title: L4 (Sig, PWR) (Scale 1:1) | File Name: CanSat_Devboard.kicad_pcb | Designer: Bertalan Nagy | Date: 2024-04-13 | Revision: + (Unreleased) |
| Sheet Path: | | Reviewer: | Size: **A4** | Sheet: **10** of **10** |